Abstract linear algebra homework6

Fisher Yu

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Problem 1

Statement. Let V be a vector space over a field $\mathbb{F}$, and let $T : V \to V$ be a linear map. Let W be an invariant subspace of T (i.e. $T(W) \subseteq W$). Prove that the assignment T' given by

$$T'(v + W) = T(v) + W$$

is a well-defined linear map $V/W \to V/W$.

Proof. **Well-definedness.** Suppose $v, v' \in V$ are such that $v + W = v' + W$ as cosets of W in V. By definition of coset equality, this holds if and only if $v - v' \in W$. We must show that $T(v) + W = T(v') + W$, i.e., that $T(v) - T(v') \in W$. Since T is linear,

$$T(v) - T(v') = T(v - v').$$

By hypothesis $v - v' \in W$, and since W is an invariant subspace of T (meaning $T(w) \in W$ for every $w \in W$), we conclude

$$T(v - v') \in W.$$

Hence $T(v) - T(v') \in W$, so $T(v) + W = T(v') + W$. This shows that T' does not depend on the choice of representative v of the coset $v + W$, so T' is well-defined.

Linearity.

Additivity. For $u, v \in V$,

$$T'\big((v + W) + (u + W)\big) = T'\big((v + u) + W\big) = T(v + u) + W = \big(T(v) + T(u)\big) + W$$

$$= \big(T(v) + W\big) + \big(T(u) + W\big) = T'(v + W) + T'(u + W),$$

where the third equality uses additivity of T.

Homogeneity. For $\lambda \in \mathbb{F}$ and $v \in V$,

$$T'\big(\lambda(v+W)\big) = T'\big((\lambda v)+W\big) = T(\lambda v)+W = \lambda\, T(v)+W = \lambda\big(T(v)+W\big) = \lambda\, T'(v+W),$$

where the third equality uses homogeneity of T.

Since T' is well-defined, additive, and homogeneous, $T' : V/W \to V/W$ is a well-defined linear map. □

Problem 2

Consider the Fibonacci sequence $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$, with $a_0 = a_1 = 1$.

(a)

Statement. For each $n \geq 1$,

$$\begin{pmatrix} a_{n-1} \\ a_n \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^{n-1} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}.$$

Proof. Let $A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$. We induct on n.

Base case ($n = 1$). We have $A^0 = I$, and

$$I \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} a_0 \\ a_1 \end{pmatrix},$$

which is exactly $\begin{pmatrix} a_{1-1} \\ a_1 \end{pmatrix}$. So the base case holds.

Inductive step. Suppose the claim holds for some $n \geq 1$, i.e.,

$$\begin{pmatrix} a_{n-1} \\ a_n \end{pmatrix} = A^{n-1} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}.$$

Multiplying both sides on the left by A:

$$A^n \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = A \begin{pmatrix} a_{n-1} \\ a_n \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a_{n-1} \\ a_n \end{pmatrix} = \begin{pmatrix} a_n \\ a_{n-1} + a_n \end{pmatrix}.$$

By the recurrence relation, $a_{n-1} + a_n = a_{n+1}$. So

$$A^n \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} a_n \\ a_{n+1} \end{pmatrix} = \begin{pmatrix} a_{(n+1)-1} \\ a_{n+1} \end{pmatrix},$$

which is the claim for $n + 1$. By induction, the formula holds for all $n \geq 1$. □

(b)

Statement. For all $n \geq 1$,

$$a_n = \frac{5 + \sqrt{5}}{10} \left(\frac{1 + \sqrt{5}}{2} \right)^n + \frac{5 - \sqrt{5}}{10} \left(\frac{1 - \sqrt{5}}{2} \right)^n.$$

Proof. Let $A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ as in part (a). We diagonalize A.

Eigenvalues. The characteristic polynomial of A is

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & 1 \\ 1 & 1 - \lambda \end{pmatrix} = -\lambda(1 - \lambda) - 1 = \lambda^2 - \lambda - 1.$$

Its roots are

$$\varphi = \frac{1 + \sqrt{5}}{2}, \qquad \psi = \frac{1 - \sqrt{5}}{2}.$$

These are distinct real numbers, so A is diagonalizable over $\mathbb{R}$.

Eigenvectors. For an eigenvalue $\lambda \in \{\varphi, \psi\}$, solving $(A - \lambda I)v = 0$ for $v = (v_1, v_2)^T$ gives $-\lambda v_1 + v_2 = 0$, i.e. $v_2 = \lambda v_1$. So $(1, \lambda)^T$ is an eigenvector for λ.

Let

$$P = \begin{pmatrix} 1 & 1 \\ \varphi & \psi \end{pmatrix}, \qquad D = \begin{pmatrix} \varphi & 0 \\ 0 & \psi \end{pmatrix},$$

so that $A = PDP^{-1}$ and hence $A^{n-1} = PD^{n-1}P^{-1}$.

Computing P^{-1}. We have $\det P = \psi - \varphi = -\sqrt{5}$, so

$$P^{-1} = \frac{1}{-\sqrt{5}} \begin{pmatrix} \psi & -1 \\ -\varphi & 1 \end{pmatrix}.$$

Applying to the initial vector.

By part (a), $\begin{pmatrix} a_{n-1} \\ a_n \end{pmatrix} = A^{n-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = PD^{n-1}P^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. First,

$$P^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{-\sqrt{5}} \begin{pmatrix} \psi - 1 \\ 1 - \varphi \end{pmatrix}.$$

Since $\psi - 1 = \dfrac{1 - \sqrt{5}}{2} - 1 = \dfrac{-1 - \sqrt{5}}{2} = -\varphi$ and $1 - \varphi = \dfrac{1 - \sqrt{5}}{2} = \psi$, this simplifies to

$$P^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{-\sqrt{5}} \begin{pmatrix} -\varphi \\ \psi \end{pmatrix} = \begin{pmatrix} \varphi/\sqrt{5} \\ -\psi/\sqrt{5} \end{pmatrix}.$$

Set $c_1 = \varphi/\sqrt{5}$ and $c_2 = -\psi/\sqrt{5}$. Then

$$D^{n-1}P^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} c_1\varphi^{n-1} \\ c_2\psi^{n-1} \end{pmatrix},$$

and therefore

$$\begin{pmatrix} a_{n-1} \\ a_n \end{pmatrix} = P \begin{pmatrix} c_1\varphi^{n-1} \\ c_2\psi^{n-1} \end{pmatrix} = \begin{pmatrix} c_1\varphi^{n-1} + c_2\psi^{n-1} \\ c_1\varphi^n + c_2\psi^n \end{pmatrix}.$$

Reading off the second coordinate,

$$a_n = c_1\varphi^n + c_2\psi^n = \frac{\varphi}{\sqrt{5}}\varphi^n - \frac{\psi}{\sqrt{5}}\psi^n.$$

Simplifying the coefficients. We claim $\dfrac{\varphi}{\sqrt{5}} = \dfrac{5 + \sqrt{5}}{10}$ and $-\dfrac{\psi}{\sqrt{5}} = \dfrac{5 - \sqrt{5}}{10}$. Indeed,

$$\frac{\varphi}{\sqrt{5}} = \frac{1 + \sqrt{5}}{2\sqrt{5}} = \frac{(1 + \sqrt{5})\sqrt{5}}{2 \cdot 5} = \frac{\sqrt{5} + 5}{10} = \frac{5 + \sqrt{5}}{10},$$

and similarly

$$-\frac{\psi}{\sqrt{5}} = -\frac{1 - \sqrt{5}}{2\sqrt{5}} = \frac{(\sqrt{5} - 1)\sqrt{5}}{10} = \frac{5 - \sqrt{5}}{10}.$$

Substituting, we obtain

$$a_n = \frac{5 + \sqrt{5}}{10}\varphi^n + \frac{5 - \sqrt{5}}{10}\psi^n = \frac{5 + \sqrt{5}}{10}\left(\frac{1 + \sqrt{5}}{2}\right)^n + \frac{5 - \sqrt{5}}{10}\left(\frac{1 - \sqrt{5}}{2}\right)^n,$$

as desired. (As a sanity check: at $n = 1$ this expression evaluates to $1 = a_1$, and at $n = 2$ it evaluates to $2 = a_2$.) $\square$

Problem 3

(a)

Statement. Let V be a vector space over $\mathbb{C}$, and let $T : V \to V$ be a linear operator such that $T^k = Id_V$ for some $k \geq 2$. Prove that the eigenvalues of T are among the kth roots of unity in $\mathbb{C}$ (of which there are exactly k, all distinct), and that T is diagonalizable.

Proof. **Step 1: the kth roots of unity are k distinct numbers.** Let $\omega = e^{2\pi i/k}$. The polynomial $x^k - 1$ factors over $\mathbb{C}$ as

$$x^k - 1 = \prod_{j=0}^{k-1}(x - \omega^j).$$

These k roots $1, \omega, \omega^2, \ldots, \omega^{k-1}$ are pairwise distinct: if $\omega^j = \omega^{j'}$ for $0 \leq j < j' \leq k-1$, then $\omega^{j'-j} = 1$ with $0 < j' - j < k$, contradicting that ω is a primitive kth root of unity. Moreover $x^k - 1$ is *squarefree*: its derivative kx^{k-1} vanishes only at $x = 0$, and 0 is not a root of $x^k - 1$ (since $-1 \neq 0$), so $x^k - 1$ and its derivative share no common root, hence $x^k - 1$ has no repeated roots.

Step 2: every eigenvalue of T is a kth root of unity. Suppose λ is an eigenvalue of T, with eigenvector $v \neq 0$, so $Tv = \lambda v$. Applying T repeatedly,

$$v = Id_V(v) = T^k(v) = \lambda^k v.$$

Hence $(\lambda^k - 1)v = 0$, and since $v \neq 0$, we conclude $\lambda^k = 1$. So λ is one of the k numbers $\omega^0, \ldots, \omega^{k-1}$ found in Step 1.

Step 3: T is diagonalizable. Since $T^k = Id_V$, the polynomial $x^k - 1$ satisfies $p(T) = T^k - Id_V = 0$, so the minimal polynomial $m(x)$ of T divides $x^k - 1$. By Step 1, $x^k - 1$ has no repeated roots, so $m(x)$, being a divisor of a squarefree polynomial, also has no repeated roots. Since $\mathbb{C}$ is algebraically closed, $m(x)$ splits into distinct linear factors over $\mathbb{C}$. By the standard criterion for diagonalizability (an operator on a finite-dimensional complex vector space is diagonalizable if and only if its minimal polynomial splits into distinct linear factors), T is diagonalizable.

Combining Steps 2 and 3: T is diagonalizable, and its (at most k, pairwise distinct) eigenvalues all lie among the k distinct kth roots of unity $e^{2\pi in/k}$, $0 \leq n \leq k-1$. □

(b)

Statement. Is the result of part (a) true when $\mathbb{C}$ is replaced by an arbitrary field?

Answer: No. The key step that fails over an arbitrary field F is Step 1 of part (a): the polynomial $x^k - 1$ need not be squarefree, and F (or its algebraic closure) need not contain k distinct kth roots of unity.

Counterexample. Let $F = \mathbb{F}_p$ (the field with p elements, p prime), let $k = p \geq 2$, and let $V = F^2$. Define $T : V \to V$ by the matrix

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

By the binomial theorem (valid since F has characteristic p, and entries of T commute),

$$T^p = \begin{pmatrix} 1 & p \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = Id_V,$$

since $p = 0$ in F. So $T^k = Id_V$ with $k = p$.

However, T has only one eigenvalue: the characteristic polynomial of T is $(x-1)^2$, and one checks directly that the eigenspace for eigenvalue 1 is one-dimensional (spanned by $(1,0)^T$), so T is *not* diagonalizable. In particular T does not have p distinct eigenvalues among the pth roots of unity in F; it has only the single eigenvalue 1. This reflects the fact that in characteristic p, the polynomial $x^p - 1$ factors as

$$x^p - 1 = (x-1)^p$$

(the "Frobenius" factorization), so it has a single root, 1, of multiplicity p rather than p distinct roots.

Thus the conclusion of part (a) — that the minimal polynomial has distinct roots and T is diagonalizable — can fail over an arbitrary field, so the result of part (a) does not hold in general. □

Problem 15

Statement. Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbb{F}$. Show that λ is an eigenvalue of T if and only if λ is an eigenvalue of the dual operator $T' \in \mathcal{L}(V')$.

Proof. Recall two basic facts about the dual map, which we use freely below.

(i) For $S \in \mathcal{L}(V,W)$, the dual map $S' \in \mathcal{L}(W',V')$ is defined by $S'(\varphi) = \varphi \circ S$ for $\varphi \in W'$. The operation $S \mapsto S'$ is linear, and ${I_V}' = I_{V'}$ (the dual of the identity is the identity). Consequently, for $S = T - \lambda I_V \in \mathcal{L}(V)$,

$$S' = (T - \lambda I_V)' = T' - \lambda {I_V}' = T' - \lambda I_{V'}.$$

(ii) Directly from the definition of the dual map, $\operatorname{null}(S') = (\operatorname{range} S)^0$: indeed $\varphi \in \operatorname{null}(S')$ means $\varphi \circ S = 0$, i.e., φ vanishes on $\operatorname{range} S$, i.e., $\varphi \in (\operatorname{range} S)^0$. (This holds for any $S \in \mathcal{L}(V,W)$, with no finite-dimensionality needed.)

(iii) If V is finite-dimensional and U is a subspace of V, then $\dim U^0 = \dim V - \dim U$. In particular, for a subspace $U \subseteq V$ of a finite-dimensional space, $U^0 = \{0\}$ if and only if $U = V$.

Now let $S = T - \lambda I_V \in \mathcal{L}(V)$, so that (by (i)) $S' = T' - \lambda I_{V'} \in \mathcal{L}(V')$.
(⇐) direction, via a chain of equivalences. We show

$$\lambda \text{ is an eigenvalue of } T' \iff \lambda \text{ is an eigenvalue of } T,$$

by tracking when each side *fails*.

By definition, λ is *not* an eigenvalue of T' if and only if $\operatorname{null}(S') = \{0\}$. By (ii), $\operatorname{null}(S') = (\operatorname{range} S)^0$, so

$$\operatorname{null}(S') = \{0\} \iff (\operatorname{range} S)^0 = \{0\} \iff \operatorname{range} S = V,$$

where the last equivalence is (iii) applied to the subspace range S of the finite-dimensional space V.

Since $S = T - \lambda I_V$ is an operator on the finite-dimensional space V (i.e., a linear map from V to itself), the rank and nullity th'm gives $\dim V = \dim \operatorname{null} S + \dim \operatorname{range} S$, and hence

$$\operatorname{range} S = V \iff \dim \operatorname{range} S = \dim V \iff \dim \operatorname{null} S = 0 \iff \operatorname{null} S = \{0\}.$$

But $\operatorname{null} S = \{0\}$ is exactly the statement that λ is *not* an eigenvalue of T (since $S = T - \lambda I_V$ and λ is an eigenvalue of T precisely when $T - \lambda I_V$ is not injective, i.e. has nonzero null space).

Chaining these equivalences:

$$\lambda \text{ not an eigenvalue of } T' \iff \operatorname{null}(S') = \{0\} \iff \text{range}$$

$$S = V \iff \operatorname{null} S = \{0\} \iff \lambda \text{ not an eigenvalue of } T.$$

Taking the contrapositive of this "iff" statement (i.e., negating both sides) gives

$$\lambda \text{ is an eigenvalue of } T' \iff \lambda \text{ is an eigenvalue of } T,$$

□

Check-In

Finished!